

Brain Death Determination (BD/DNC)

AAN/AAP/CNS/SCCM 2023 Pediatric & Adult Consensus Guideline Reference.

1. PREREQUISITES

Irreversible Coma
Temp $\geq 36^{\circ}\text{C}$ | Hemodyn OK

2. CLINICAL EXAM

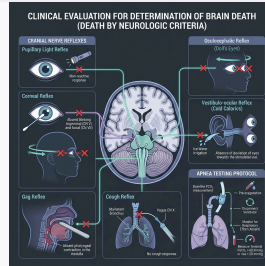
No Motor Response
Absent Brainstem Reflexes

3. APNEA TESTING

Absence of Resp Drive
PaCO₂ ≥ 60 & pH < 7.30

4. ANCILLARY TESTING

Only if exam or apnea test
cannot be completed/concluded



1. Prerequisites & Stability

- **Etiology:** Known, irreversible, catastrophic brain injury.
- **Core Temp:** $\geq 36.0^{\circ}\text{C}$ (96.8°F). Warm if hypothermic.
- **Hemodynamics:**
 - Adults: SBP ≥ 100 mmHg (or MAP ≥ 60 mmHg)
 - Pediatrics (Age-specific SBP):
 - Term newborn to 30 days: ≥ 60 mmHg
 - Infants 31 days to 1 year: ≥ 70 mmHg
 - Children 1 to 10 years: $\geq 70 + (2 \times \text{age})$ mmHg
 - Adolescents > 10 years: ≥ 90 mmHg
- **Exclusions:** Exclude CNS depressants (≥ 5 half-lives) and neuromuscular blockade (TOF 4/4 twitch present).
- **Metabolic:** Correct severe endocrine or electrolyte derangements.

2. Neurological Examination

- **Coma:** Complete absence of arousal. No motor responses to pain (spinal reflexes like triple flexion allowed).
- **Pupils:** Mid-sized/dilated (4–9 mm), completely unresponsive to intense light bilaterally.
- **Corneal Reflex:** No blink to cotton swab contact.
- **Oculocephalic (Doll's Eyes):** No eye deviation with head turn (ensure C-spine cleared).
- **Oculovestibular (Cold Calorics):** No eye movement for ≥ 1 min after 50 mL ice-water irrigation (confirm tympanic membrane intact).
- **Gag & Cough:** Absent gag (pharyngeal stim) and cough (tracheal suction stim).
- **Facial Motor:** No grimace to TMJ or supraorbital pressure.

3. Apnea Testing Protocol

- **Preparation:** Pre-oxygenate with 100% O₂ for ≥ 10 mins. Baseline arterial pCO₂ must be 35–45 mmHg, pH 7.35–7.45.
- **Procedure:** Deliver passive O₂ via CPAP (preferred, 100% O₂ at 4–6 L/min) to maintain oxygenation while ventilator is paused.
- **ECMO Tip:** Decrease sweep gas flow to 0.5–1.0 L/min with 100% O₂; monitor closely for any chest rise.
- **Observation:** Monitor for spontaneous respiratory effort for 8–10 mins.
- **Target:** Final pCO₂ ≥ 60 mmHg AND ≥ 20 mmHg above baseline, with pH < 7.30 .

Abort/Stop Criteria: Abruptly stop and draw ABG if:

- SBP < 100 mmHg or MAP < 60 mmHg.
- SpO₂ $< 85\%$ for > 30 seconds.
- New significant cardiac arrhythmias occur.

4. Ancillary Testing Guidelines

Used **only** when clinical exam or apnea test cannot be completed safely (e.g. severe hypoxemia, facial trauma).

ACCEPTED TESTS

- **Conventional 4-Vessel DSA:** Confirms absence of flow.
- **SPECT Perfusion:** Shows absence of cerebral uptake.
- **TCD:** (Transcranial Doppler) **Adults only**; must show oscillating flow or spikes.

UNACCEPTABLE TESTS

- **EEG: NO LONGER ACCEPTED** (cannot evaluate brainstem).
- **CTA: NOT ACCEPTABLE** (insufficient validation).
- MRI/MRA: Not accepted.

BD/DNC DOCUMENTATION & REPEAT EXAM REQUIREMENTS

- ✓ **Adults:** One clinical exam and apnea test is sufficient (per 2023 AAN).
- ✓ **Rewarming Period:** Wait ≥ 24 hours after rewarming from therapeutic hypothermia before testing.
- ✓ **Pediatrics:** Two separate exams & apnea tests required (intervals: 24h for newborns; 12h for infants/children).
- ✓ **Time of Death:** Recorded when pCO₂ meets target or when the ancillary test result is officially read.

Consensus Guideline: Greer DM, et al. Pediatric and Adult Brain Death/Death by Neurologic Criteria Consensus Practice Guideline. *Neurology*. 2023;101(24):1112–1132. PMID: 37827878.

Ancillary Update: Wijdevits EF, et al. Practice parameter update: determining brain death in adults. *Neurology*. 2010;74(23):1911–1918. PMID: 20530327.